

Used SQL Queries in BigQuery

1. Retail Market Analysis

Note: Data was named: ` sales `

```
WITH country_totals AS (  
  SELECT  
    r.Country,  
    COUNT(DISTINCT s.`Retailer code`) AS Total_Retailers,  
    SUM(s.Quantity * s.`Unit sale price`) AS Current_Volume  
  FROM `sales.sales` AS s  
  JOIN `sales.retailers` AS r ON s.`Retailer code` = r.`Retailer code`  
  GROUP BY r.Country  
)  
  
base_sales AS (  
  SELECT  
    r.Country,  
    r.`Retailer code`,  
    SUM(s.Quantity * s.`Unit sale price`) AS retailer_revenue  
  FROM `sales.sales` AS s  
  JOIN `sales.retailers` AS r ON s.`Retailer code` = r.`Retailer code`  
  GROUP BY 1, 2  
)  
  
ranked_sales AS (  
  SELECT  
    Country,  
    retailer_revenue,  
    ROW_NUMBER() OVER(PARTITION BY Country ORDER BY retailer_revenue DESC) AS rnk  
  FROM base_sales  
)  
  
market_share_metric AS (  
  SELECT  
    Country,  
    SUM(CASE WHEN rnk <= 3 THEN retailer_revenue ELSE 0 END) AS top_3_revenue  
  FROM ranked_sales  
  GROUP BY Country  
)  
  
SELECT  
  ct.Country,  
  CASE WHEN (m.top_3_revenue / NULLIF(ct.Current_Volume, 0)) >= 0.80 THEN 'Big-player market' ELSE  
'Competitive market' END AS Market_Type,  
  -- Updated here: Removed spaces and backticks to use a clean underscore alias  
  ROUND((m.top_3_revenue / NULLIF(ct.Current_Volume, 0)) * 100, 2) AS Top_3_share,  
  ct.Total_Retailers,  
  -- Competitive market: Dustin wants a 15% increase in the number of retailers (1.15)
```

```

CASE WHEN (m.top_3_revenue / NULLIF(ct.Current_Volume, 0)) < 0.80 THEN CEIL(ct.Total_Retailers *
1.15) ELSE NULL END AS Target_Retailers,
ROUND(ct.Current_Volume, 2) AS Current_Volume,
-- Big-player market: Dustin wants a 10% increase in sales volume (1.10)
CASE WHEN (m.top_3_revenue / NULLIF(ct.Current_Volume, 0)) >= 0.80 THEN
ROUND(ct.Current_Volume * 1.10, 2) ELSE NULL END AS Target_Volume
FROM country_totals AS ct
JOIN market_share_metric AS m ON ct.Country = m.Country
ORDER BY Top_3_share DESC;

```

2. Order Method Analysis

Note: Data was named: `sales`

```

SELECT
  s.Date,
  r.Country,
  m.`Order Method Type` AS Order_Method,
  (s.quantity * s.`Unit sale price`) AS Revenue,
  (s.quantity * (s.`Unit sale price` - p.`Unit cost`)) AS Profit,
  s.quantity AS Quantity
FROM `sales.sales` AS s
INNER JOIN `sales.products` AS p
  ON s.`product number` = p.`product number`
INNER JOIN `sales.methods` AS m
  ON s.`Order method code` = m.`Order method code`
INNER JOIN `sales.retailers` AS r
  ON s.`Retailer Code` = r.`Retailer Code`;

```

```

SELECT
  *
FROM `sales.sales` AS s
INNER JOIN `sales.products` AS p
  ON s.`product number` = p.`product number`
INNER JOIN `sales.methods` AS m
  ON s.`Order method code` = m.`Order method code`
INNER JOIN `sales.retailers` AS r
  ON s.`Retailer Code` = r.`Retailer Code`;

```